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Batch 15

Algorithm

Main Algorithm — Menu-Driven String Operations

1. Start.
2. Declare an integer variable **choice**.
3. Display the menu:
 - Convert lowercase string to uppercase.
 - Reverse a string.
 - Check whether a string is a palindrome.
 - Exit.
4. Read **choice** from the user.
5. Use a **switch** statement:
 - If **choice** = 1, call **toUppercase()**.
 - If **choice** = 2, call **reverseString()**.
 - If **choice** = 3, call **checkPalindrome()**.
 - If **choice** = 4, display an exit message.
 - Otherwise, display "**Invalid choice**".
6. Repeat Steps 3–5 until the user selects 4.
7. Stop.

Algorithm for toUppercase()

1. Start the function.
2. Declare a character array **str** and integer **i**.
3. Read the string from the user.
4. Set **i** = 0.
5. Repeat while **str[i] != '\0'**:
 - Check whether **str[i]** lies between 'a' and 'z'.
 - If yes, convert it to uppercase using:
str[i] = str[i] - ('a' - 'A');
 - Increment **i**.
6. Display the uppercase string.
7. Return to the main function.

Algorithm for reverseString()

1. Start the function.
2. Declare a character array **str** and integers **i** and **length**.

3. Read the string from the user.
4. Set `length = 0`.
5. Traverse the string until '`\0`' is encountered and calculate its length.
6. Set `i = length - 1`.
7. Print the characters from `str[i]` to `str[0]` by decreasing `i`.
8. Display the reversed string.
9. Return to the main function.

Algorithm for **checkPalindrome()**

1. Start the function.
2. Declare a character array `str`, integers `i`, `length`, and set `palindrome = 1`.
3. Read the string from the user.
4. Calculate the length of the string manually.
5. Compare the first character with the last character, the second with the second-last, and so on.
6. If any pair does not match:
 - Set `palindrome = 0`.
 - Stop comparison.
7. If `palindrome = 1`, display that the string is a palindrome.
8. Otherwise, display that the string is not a palindrome.
9. Return to the main function.

Test Case Table

Test Case	Menu Choice	Input	Expected Output	Purpose
1	1	hello	HELLO	Normal lowercase
2	1	computer science	COMPUTER SCIENCE	Lowercase string containing a space
3	1	abc123	ABC123	String containing letters
4	1	a	A	Single lowercase
5	2	hello	olleh	Normal string reversal
6	2	computer	retupmoc	Longer string reversal
7	2	abc 123	321 cba	String containing space
8	2	A	A	Single-character
9	3	madam	madam is a palindrome.	Odd-length palindrome
10	3	noon	noon is a palindrome.	Even-length
11	3	hello	hello is not a	Non-palindrome
12	3	a	a is a palindrome.	Single-character
13	5	—	Invalid choice! Please enter 1 to 4.	Invalid menu choice
14	4	—	Exiting program...	Exit operation

C program

```
#include <stdio.h>

void toUppercase(void);
void reverseString(void);
void checkPalindrome(void);

int main()
{
    int choice;

    do
    {
        printf("\n==== STRING OPERATIONS =====\n");
        printf("1. Convert lowercase string to uppercase\n");
        printf("2. Reverse a string\n");
        printf("3. Check if a string is a palindrome\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice)
        {
            case 1:
                toUppercase();
                break;

            case 2:
                reverseString();
                break;

            case 3:
                checkPalindrome();
                break;

            case 4:
                printf("Exiting program...\n");
                break;

            default:
                printf("Invalid choice! Please enter 1 to
4.\n");

```

```

    }

    } while (choice != 4);

    return 0;
}

/* Function to convert lowercase string to uppercase */
void toUppercase(void)
{
    char str[100];
    int i;

    printf("Enter a lowercase string: ");
    scanf(" %99[^\n]", str);

    for (i = 0; str[i] != '\0'; i++)
    {
        if (str[i] >= 'a' && str[i] <= 'z')
        {
            str[i] = str[i] - ('a' - 'A');
        }
    }

    printf("Uppercase string: %s\n", str);
}

/* Function to reverse a string */
void reverseString(void)
{
    char str[100];
    int i, length = 0;

    printf("Enter a string: ");
    scanf(" %99[^\n]", str);

    /* Find length without strlen() */
    while (str[length] != '\0')
    {
        length++;
    }

    printf("Reversed string: ");

    for (i = length - 1; i >= 0; i--)

```

```

    {
        printf("%c", str[i]);
    }

    printf("\n");
}

/* Function to check whether a string is a palindrome */
void checkPalindrome(void)
{
    char str[100];
    int i, length = 0;
    int palindrome = 1;

    printf("Enter a string: ");
    scanf(" %99[^\n]", str);

    /* Find length without strlen() */
    while (str[length] != '\0')
    {
        length++;
    }

    for (i = 0; i < length / 2; i++)
    {
        if (str[i] != str[length - 1 - i])
        {
            palindrome = 0;
            break;
        }
    }

    if (palindrome)
        printf("%s is a palindrome.\n", str);
    else
        printf("%s is not a palindrome.\n", str);
}

```

Output

```
Last login: Tue Sep 29 22:35:23 on ttys000
bhavya@Bhavyas-MacBook-Air ~ % nano day37.c
bhavya@Bhavyas-MacBook-Air ~ % clang day37.c -o day37
bhavya@Bhavyas-MacBook-Air ~ % ./day37
```

```
===== STRING OPERATIONS =====
```

1. Convert lowercase string to uppercase
2. Reverse a string
3. Check if a string is a palindrome
4. Exit

Enter your choice: 1

Enter a lowercase string: hello world

Uppercase string: HELLO WORLD

```
===== STRING OPERATIONS =====
```

1. Convert lowercase string to uppercase
2. Reverse a string
3. Check if a string is a palindrome
4. Exit

Enter your choice: 2

Enter a string: computer

Reversed string: retupmoc

```
===== STRING OPERATIONS =====
```

1. Convert lowercase string to uppercase
2. Reverse a string
3. Check if a string is a palindrome
4. Exit

Enter your choice: 3

Enter a string: madam

madam is a palindrome.

```
===== STRING OPERATIONS =====
```

1. Convert lowercase string to uppercase
2. Reverse a string
3. Check if a string is a palindrome
4. Exit

Enter your choice: 4

Exiting program...

bhavya@Bhavyas-MacBook-Air ~ % █